

VOCALIZATIONS OF THE BLACK-TAILED PRAIRIE DOG, *CYNOMYS LUDOVICIANUS*

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Abstract. Seven variable vocal displays of *Cynomys ludovicianus* were studied in a captive colony, and checked through limited field work and literature reports. The form of each display makes available information identifying its user. In addition, since each display correlates probabilistically with a consistent set of activities, it makes available information about the behaviour in which the communicator is, or may become, engaged. At least five different kinds of behaviour can be specified through the use of these displays.

In an earlier paper we described the patterns of social organization found in the black-tailed prairie dog (*Cynomys ludovicianus*) colony of the Philadelphia Zoo, most of which are comparable to patterns typical of wild colonies (Smith et al. 1973). We provided details that are pertinent to the study of display behaviour; for example, details on prominent patterns of social behaviour such as boundary 'challenging', territories, bonds, etc., and also zoo population composition and density.

This paper describes the forms and uses of the vocal displays of black-tailed prairie dogs. It is based on observations made of the zoo colony between 1965 and 1975, and is augmented by observations of the wild colony at Devil's Tower, Wyoming (brief in July 1968, and much more extensive in May and June of 1970 and 1971), and by reference to the literature. Quantitative analyses are made of data obtained in the initial 3 years of the zoo study, from autumn 1965 to summer 1968.

Analyses of the patterns of use of the vocalizations determine what information each makes available about the behaviour of the communicator. When analysed in terms of component 'messages', this information appears to be of kinds that are widespread among diverse species of animals (see W. J. Smith 1969).

Barks and Barking Patterns

Description

Barks are the brief vocalizations that led this ground squirrel to be called a 'dog'. Black-tailed

prairie dogs have two very similar forms of barking: one is relatively more prolonged than the other (Fig. 1(a), (b)). Within each form there is variation in peak frequency, duration, and 'harshness' (see below), but the two forms do not usually intergrade. An apparent intermediate has two peaks, the first being the briefer (Fig. 1(c), (d)).

Barks were repeated much more often than uttered singly, but might be repeated discontinuously or in series that tended to have regular intervals between successive barks. We termed the latter 'continuous barking', but could find no simple criterion for distinguishing bouts of it from discontinuously repeated barks. There was, for instance, no fixed interval for continuous barking: bouts commonly had from as few as 10 or 20 barks per minute to as many as one or two per second. Occasional bouts reached four barks per second, and very rarely even this rate was exceeded. As an arbitrary lower limit, we chose to recognize any sequence of three or more barks not separated by intervals of more than a minute as a bout of continuous barking, although this probably led us to include in our category some barks that would have been more accurately classified as discontinuously uttered. Recognition of continuous bouts was further complicated because barking sequences often begin with slightly irregular intervals, falter or change pace at times, and may terminate gradually.

Tallies of bouts of continuous barking are unlike tallies of discontinuously uttered barks in that a bout may have a considerable duration and include many barks. Most continuous bouts lasted 1 or 2 min. However, some exceeded 5 min. In the second and third years of the study there were 32 such long bouts, representing 11 per cent of the 284 bouts tallied, and they

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totalled approximately 300 to 330 min, an average of about ten minutes apiece.

Barks grouped into brief bursts with a distinctive 'chitter' sound were recognized as a third kind of display. Chitter barking bursts typically comprise four to eight component barks, the first a fairly typical sharply peaked or brief bi-peaked form, and the remainder briefer and modified in harmonic structure (Fig. 2(a)). The components are very rarely of the broader bark form (Fig. 2(c)), except perhaps when used by pups, from which we obtained good recordings only in 1971. Occasionally chitter barking is harsh (Fig. 2(b)).

By dividing black-tailed prairie dog vocalizations into 'barks' and two patterns of barking our scheme departs from previously published descriptions, none of which recognized single or discontinuously uttered barks as distinctive. Waring's scheme (1970) is otherwise closest, recognizing 'repetitious barks' of either one or two syllables, and transitional forms; his one-syllable units are our broadly peaked barks, and all others correspond to our bi-peaked forms. His 'chatter barks' correspond to most of our chitter barking, although he did not use a 'burst pattern' criterion. His 'chuckle' appears to correspond to some of our narrowly peaked

continuous bi-peaked chitter recognized 'warning', and our continuous we found no 'warning' and though both they could have opportunity to analyses was

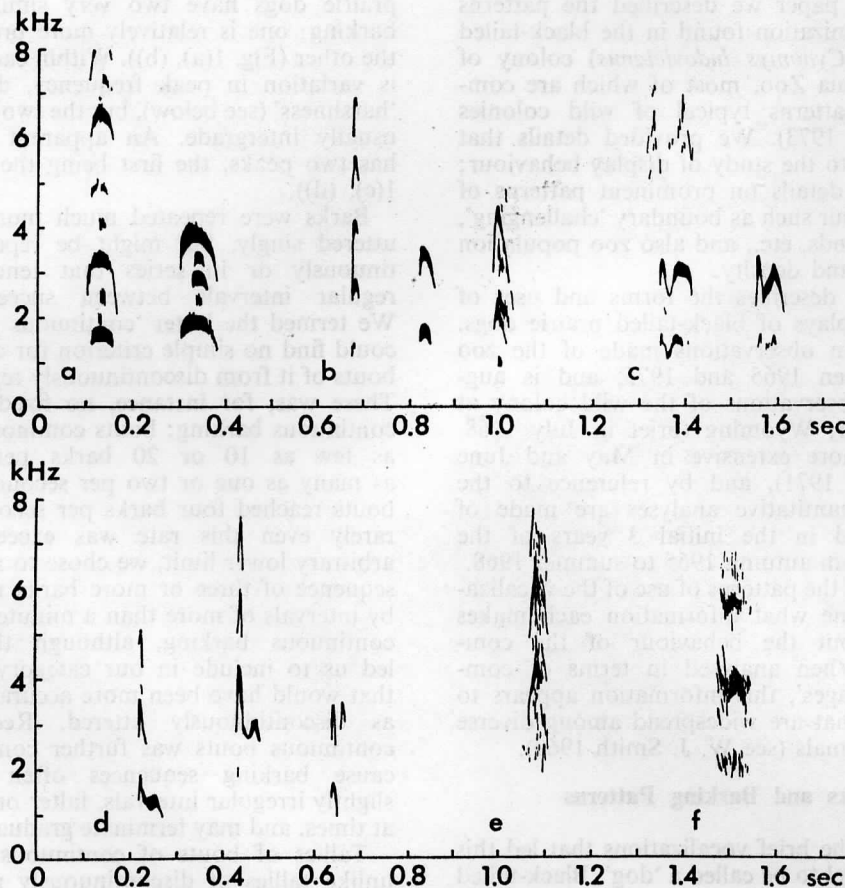


Fig. 1. The bark vocalization: (a) two separate examples of the 'broad' form, (b) three separate examples of the 'narrow' form, (c) two separate bi-peaked examples, and (d) three separate examples in which the first peak is the more prominent. Harsh single (e) and bi-peaked (f) barks show vertical lines that are typical of noise; these are very hard to trace in fine detail, with the result that the figure is slightly diagrammatic.

continuous barking and perhaps to narrowly peaked chitter barking. King (1955, pp 71-73) recognized 'warning', 'hawk warning', 'defensive', and 'muffled' barking, all overlapping our continuous barking category. Like Waring, we found no consistent differences between 'warning' and 'hawk warning' categories, although both King and R. E. Smith (1958) felt they could hear consistent differences. (The opportunity to tape-record and do sonagraphic analyses was not readily available to King or

R. E. Smith, making it difficult to recognize details of form.) King's last two categories both include series with very irregular intervals, and therefore embrace many barks we might label as discontinuously uttered. Finally, King's 'little barks' and Anthony's 'chirr' (1955) must correspond to much of our chitter barking category. Since there is relatively little agreement among schemes, based primarily on form, we have developed a classification of these inter-

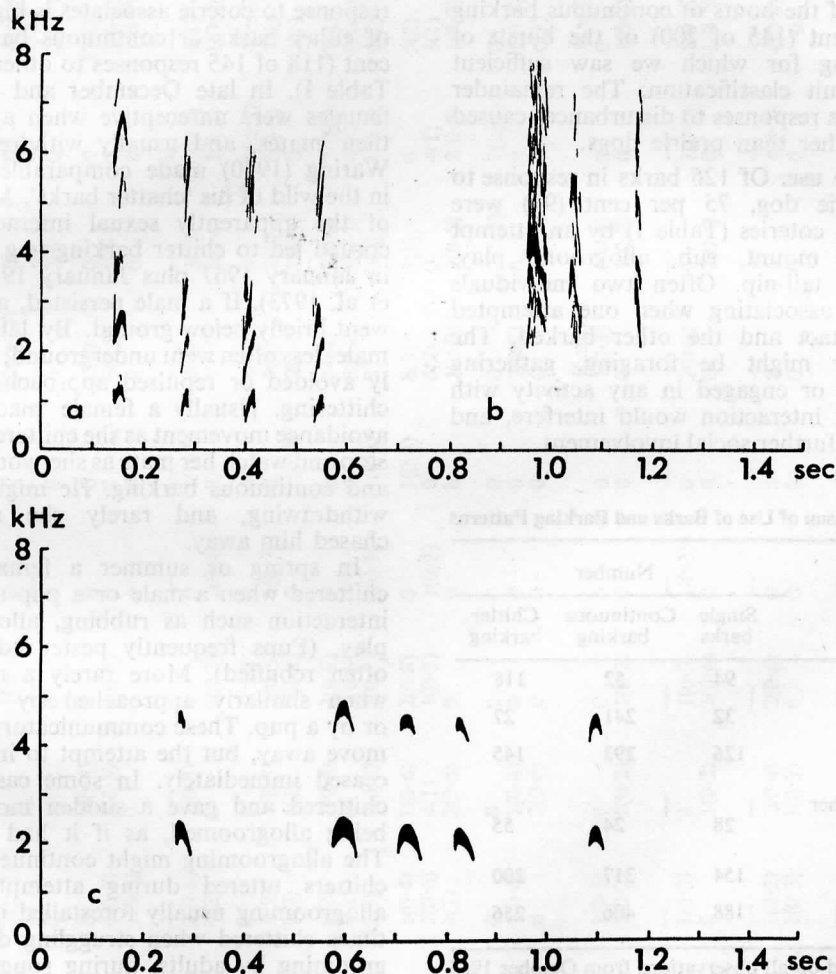


Fig. 2. A typical burst of chitter barking (a) with four components, and a harsh chitter barking burst (b) with three components. The burst with five unusually broad components (c) was recorded from a pup.

Table II. Use of Continuous Barking in Different Situations

	1*	2	3	4	5	6	7	8	U	Total No. of bouts	No. hr of obser.	Rate /hr
Oct. '65	2(≈3)	0	1(≈1)	0	0	12(≈17)	0	0	4(≈6)	19	0.7	≈27
'66	0	1(0.2)	0	0	0	0	2(0.3)	0	2(0.3)	5	5.9	1
'67	0	0	0	0	0	0	0	0	0	0	0	0
Nov. '65	0	1(0.5)	0	0	0	12(6)	0	0	12(6)	25	2.2	11
'66	4(0.5)	2(0.3)	5(0.8)	0	0	6(0.8)	1(0.1)	0	3(0.4)	22	7.9	3
'67	1(0.5)	0	0	0	0	0	0	0	1(0.5)	2	4.7	1
Dec. '65	0	0	0	0	0	0	0	0	0	0	0	0
'66	2(0.6)	1(0.3)	0	0	1(0.3)	9(3)	3(1)	1(0.3)	4(1)	21	3.2	7
'67	9(2)	0	1(0.2)	2(0.4)	0	1(0.2)	2(0.4)	1(0.2)	1(0.2)	17	4.8	4
Jan. '66	0	0	1(≈1)	0	0	4(≈5)	0	1(≈1)	4(≈5)	10	0.8	≈13
'67	9(2)	1(0.3)	2(0.5)	1(0.3)	0	7(2)	7(2)	2(0.5)	7(2)	36	3.8	10
'68	9(2)	5(1)	3(0.8)	0	1(0.3)	2(0.5)	12(3)	0	5(1)	37	3.7	10
Feb. '66	6(2)	1(0.3)	3(0.8)	0	0	14(4)	2(0.6)	3(0.8)	13(4)	42	3.6	12
'67	4(1)	0	3(0.8)	4(1)	0	3(0.8)	1(0.3)	4(1)	2(0.5)	21	3.7	6
'68	12(3)	3(0.8)	1(0.3)	2(0.5)	1(0.3)	0	6(2)	1(0.3)	6(2)	32	3.7	9
Mar. '66	1(0.4)	1(0.4)	2(0.8)	0	1(0.4)	8(3)	3(1)	0	5(2)	21	2.5	8
'67	2(0.6)	4(1)	1(0.3)	1(0.3)	0	6(2)	0	4(1)	3(1)	21	3.4	6
'68	4(1)	2(0.5)	0	0	0	1(0.2)	2(0.5)	2(0.5)	1(0.2)	12	4.2	3
Apr. '66	0	0	0	0	0	0	2(≈3)	1(≈1)	0	3	0.8	≈4
'67	1(0.4)	1(0.4)	0	0	0	2(0.7)	0	0	1(0.4)	5	2.7	2
'68	0	3(0.6)	0	0	0	0	4(0.8)	2(0.4)	6(1)	15	4.9	3
May '66	1(0.3)	0	0	0	0	1(≈3)	1(0.3)	0	1(≈3)	2	0.3	≈7
'67	1(0.2)	1(0.2)	1(0.2)	0	0	1(0.3)	1(0.2)	0	0	3	3.7	1
'68	0	0	0	0	0	0	0	0	1(0.2)	5	4.4	1
June '66	0	0	0	0	0	0	0	0	0	0	0	0
'67	1(0.3)	0	1(0.3)	1(0.3)	0	2(0.6)	0	1(0.3)	1(0.3)	7	3.6	2
'68	0	1(0.2)	3(0.6)	0	0	0	2(0.4)	0	1(0.2)	7	5.2	1
July '66	0	0	0	0	0	0	0	0	0	0	0	0
'67	1(0.2)	2(0.4)	2(0.4)	0	0	0	0	1(0.2)	4(0.8)	10	5.2	2
'68	1(0.3)	1(0.3)	0	0	0	0	0	0	0	2	3.8	1
Aug. '66	0	0	0	0	0	0	0	0	0	0	0	0
'67	0	1(0.4)	0	0	0	1(0.4)	1(0.4)	0	1(0.4)	4	2.6	2
'68	0	0	0	0	0	0	0	0	0	0	2.0	0
Total	71(0.7)	32(0.3)	31(0.3)	11(0.1)	4(0.4)	92(0.9)	52(0.5)	24(0.3)	89(0.9)	406	98.0	4.1

*See notes opposite (p. 157).

TABLE II

The columns represent the following situations:

Inter-coterie use, related to boundary challenges:

(1) A male challenges a neighbouring male or female. The neighbouring female runs to a burrow, if not already at one, and begins continuous barking. Her mate may return the challenge, but she rarely does.

(2) Same, but challenge initiated by a female. If she challenges another female, the latter may return the challenge and do continuous barking primarily during pauses in the challenging.

(3) A male or (rarely) a female challenges a neighbouring male who then runs to a burrow, if not already at one, and begins continuous barking. Sometimes it returns the challenge before retiring and doing continuous barking.

(4) The communicator, usually a female, calls from a burrow while watching a neighbour who challenged her recently, and/or who is now challenging at another boundary.

(5) A female wanders into or near a neighbouring territory and is approached by an individual from it. She either runs to a burrow in her own territory and barks continuously, or returns the challenge and does continuous barking afterward.

(6) Remaining cases of challenge situations, not further categorized through lack of data.

Intra-coterie use:

(7) A male approaches his mate and may anal-sniff, kiss-greet, or fight her. She then runs to a burrow, if not already at one, and begins continuous barking. Only once were the roles reversed and a female approached her mate, who vocalized.

Responses to other species:

(8) The communicator sees other species approaching, or hears a loud noise, and runs to or remains at a burrow looking alert. It may then begin continuous barking while facing the disturbance, or may watch for some time before vocalizing.

Unclassifiable:

(U) Events which cannot be accurately classified because we have too few data.

Figures to the left in each column indicate the total number of bouts of continuous barking within the observation periods of a given month. This is converted to the number of bouts per hour, in parentheses.

Data are based on all observations from October 1965 through May 1966, and a sample of approximately half the observation periods from November 1966 through August 1968.

Dashes indicate that no observations were made.

per cent) began in response to an approaching neighbour (Table I). Usually it was a female who barked (columns 1 and 2, Table II), whether she or her mate was approached. She often fled to a burrow and began barking before the staring, tensely approaching neighbour had started a full challenge. Sometimes (column 4, Table II) a prairie dog began continuous barking while watching a neighbour who had challenged her but was not then challenging (or was challenging a member of some other coterie). In four cases (column 5, Table II) a female provoked a challenge by intruding or nearly intruding into a neighbouring territory, and responded by running to a hole in her own area and beginning continuous barking, or by first returning the challenge then doing continuous barking after the incident. Continuous barking was not done at or beyond a territorial boundary.

In only 32 (25 per cent) of the 126 barks done singly in response to another prairie dog was the communicator responding to an attempted challenge, or to a sudden lunging in incipient attack during a challenge encounter (Table I). Both sexes barked in these circumstances, females when either being challenged themselves, or watching their mates being challenged.

Only 27 (19 per cent) of the 145 chitters were in response to challenges (Table I), most in late autumn and winter by females who might then return the challenge. The few times a female chitter barked on being approached by her mate as he left a challenge were considered as intra-coterie interactions.

Responses to other species. The sight of an approaching person, a large bird flying over, a startling noise, an intruding peafowl, or a pigeon competing for food sometimes elicited barks or patterned barking from zoo prairie dogs, but only 28 barks (12 per cent of the 154 that were classifiable, Table I), 24 bouts of continuous barking (8 per cent of 317, Table I), and 55 cases of chitter barking (28 per cent of 200, Table I). A prairie dog might utter two or more brief barks, alertly watch, then run to a burrow and begin continuous barking. Or it might chitter on running, or be silent until it had run to a burrow, or until it had watched for some time, and then begin continuous barking. Sometimes one chittered as it warily faced the stimulus without fleeing.

Comparable responses were elicited in Wyoming by hawks and terrestrial predators,

persons, and birds. King (1966) assigned such 'alarm' and 'warning' and Waring (1966) 'chuckles'.

Variant Forms

(1) The 'sharp' (b), (c), (d); patterned series of sharply peaked frequencies and 'squeaky'; the 'uncommon' good material behaviour pattern, as they of fidelity recorded throughout the details of pattern the following with broadly is not much natives to interest safety of a barking and there during a shift to bip with increased further shift this trend. Barks are sharply narrowly peaked fleeing.

(2) Barks vary 'harshness' (F) when the communication draw, and so that was both harshness is descending the rasp call.

In another, the initial in frequency descended, they may correlate escape without tacking.

(3) Very infrequently, individuals. M barking pattern 'mew'; perhaps 'grunting or c

persons, and low-flying magpies or other large birds. King (1955) and R. E. Smith (1958) assigned such cases to their 'warning' (or 'alarm') and 'predator or hawk warning' barks, and Waring (1970) to 'repetitious barks' and 'chuckles'.

Variant Forms and Their Uses

(1) The 'shape' of the bark varies (Fig. 1(a), (b), (c), (d)); most barks uttered singly or in patterned series, are broadly peaked. More sharply peaked barks often reach higher frequencies and sound slightly higher, sharper, or 'squeaky'; they and bipeaked forms are not uncommon. Our techniques did not provide good material for analysis of the communicator's behaviour patterns that correlated with each form, as they were adapted for dictation instead of fidelity recording, and as we followed activity throughout the group rather than the full details of particular incidents. Nonetheless, the following trends are strongly suggested; with broadly peaked forms escape behaviour is not much more likely than are other alternatives to interacting, although escape to the near-safety of a burrow entrance often precedes barking and the communicator usually remains there during continuous barking. Within a bout, a shift to bipeaked forms appears to correlate with increased likelihood of further escape, and a further shift to sharply peaked barks continues this trend. Barks given while fleeing to a burrow are sharply peaked, and Waring's (1970) narrowly peaked 'chuckles' were uttered while fleeing.

(2) Barks vary in other ways. The addition of 'harshness' (Fig. 1(e), (f)) was often detected when the communicator resisted, did not withdraw, and sometimes attacked a prairie dog that was bothering it. The characteristic of harshness is discussed further below in considering the rasp call.

In another, apparently uncommon, modification the initial arm of a sharp chevron ascended in frequency less rapidly than the terminal arm descended, the reverse of the usual form. This may correlate with a decreased likelihood of escape without increased probability of attacking.

(3) Very faint vocalizations were detected infrequently, primarily from one or two zoo individuals. Most appeared to be variations of barking patterns, and sounded like series of 'mew'; perhaps comparable is a low pitched 'grunting or quiet coughing noise' that Waring

(1970) sometimes elicited on restraining an animal. Mews were heard only 12 times in Wyoming and have not been otherwise reported from free-ranging animals. The faintness of mews resulted in extremely poor recordings, and few could be used for even marginal sonographic analysis. The low-pitched sound is determined by generally low frequencies of maximum energy.

Whether or not mews are distinctive physically other than in being faint, their occurrence was quite distinctive. The communicator usually appeared to seek interaction, although this varied from attacking to rubbing or kiss-greeting, and its attempts always appeared to be thwarted. All of the situations were agonistic, and mews at times were interspersed with tooth chattering (see below) or jump-yips (primarily in the usage of male 9, who jump-yipped excessively, Smith et al. 1976).

The apparent restriction of mewing to a few individuals, one of them socially abnormal, suggests that it may be aberrant behaviour. On the other hand, its faintness suggests that observers may often miss mewing. Many rodents, in fact, have faint vocalizations that could be useful in interactions when animals are close together, e.g. in mothering infants or when copulating; repetitive series of faint ultrasounds have been implicated in the copulatory interactions of several rodents (reviewed by Brooks & Banks 1973, p. 69). Most such interactions may occur in subterranean tunnels and quarters.

Mewing may represent a faint and unstudied portion of the vocal repertoire. Another faint vocalization is the 'purr' that Waring's (1970) pet prairie dog uttered when scratched by a person. The sounds of tooth chattering (see below) are often faint, and at times barks, continuous barking, and yips can be very faint. Whether forms are altered in these faint versions is impossible to determine from our material.

(4) The interval between successive barks in patterned barking varies; it usually decreases as the situation becomes more extreme. For example, if an individual is barking from a burrow as a challenger approaches, the intervals decrease as the distance decreases. Toward the end of a bout, as a communicator begins to move locally, perhaps foraging, the intervals increase and may become irregular.

At times a series of barks can become very rapid (a rate of five to eight barks per second being sustained for a second or two), each bark

then being a very brief unit. Such series were recorded in events in which the communicator was being closely pressed by a challenger. In form, the series were intermediate between continuous barking and chitters, although lacking a clearly developed burst pattern. They were recorded only in the first winter of the study and in the winter of 1975, in both of which the population was unusually dense. A second sort of intermediate involved more prolonged barks (standard or bipeaked forms) in series of brief bursts of two or three units each. Rarely recorded, this series occurred in challenges when an aggressor faced the communicator immediately in front of the latter's burrow. The communicator did not challenge or challenged less vigorously, but fought if pushed further or for a long time. As the opponent moved back the communicator immediately switched to standard, broad unit continuous barking.

Messages of Barks and Barking Patterns

Bark vocalizations correlate with specific kinds of activities of the communicator. They thus make available information enabling prediction of its probable behaviour. First, there is information about interacting: it is very probable that the communicator will avoid, shun, or actively reject most attempts of other individuals to interact. It may bark and move away rapidly, often fleeing to a burrow. Sometimes it barks or chitters while running, or barks after arriving at a burrow; continuous barking is usually done from a burrow, rarely beginning while running there. When an individual does begin to interact after barking, it is usually in response to a challenge: the communicator responds by defensively rejecting a neighbour's attempts to intrude, and is ready to terminate the challenge when the other desists. By this response it may evade other kinds of interaction, such as attack.

Predators elicit a minority of barks and barking patterns in circumstances comparable to those of challenges: the communicator avoids some kinds of interaction with a predator, even while interacting with it in other ways. Although the participants are of different species, the encounters are 'interactions' in that distant predators do not simply act upon prairie dogs: prey and predator respond to each other's moves.

Second, there is the information that the communicator may choose a behavioural alternative incompatible with interacting, but no single alternative is specified. The communicator

has many alternatives at different times: to withdraw, forage, gather nest material, sun, and so forth (referred to by W. J. Smith 1969 as its 'general set' of alternatives). If alternatives all involve interacting (e.g. in challenging, and occasionally when being allogroomed) some aspects of the encounter are resisted, i.e. there is still a sense in which interacting is being resisted. Yet, although barks, continuous barking and chitter barking per se give little information about particular alternatives, different physical forms of the vocal units may provide information about agonistic selections. Escape may be more likely if the bark unit(s) are sharply peaked, less likely if not. Attack is more likely if they are harsh; e.g. after a harsh chitter it is the recipient, not the communicator, who often withdraws.

Third, information is also available from barks and continuous barking about the behaviour of being attentive. Before the communicator proceeds with its alternative to interaction it will pause and alertly monitor the stimulus animal. With continuous barking it continues to monitor at the expense of other alternatives, providing supplementary information about the stability of its attentive behaviour. Because its formal characteristics include a sustained pattern of repetition, continuous barking is what Schleidt (1973) terms 'tonic' signalling.

In chitter barking the special pattern is not tonic, but through rapid repetition yields a brief, compound unit. A chitter burst very often comes as the communicator makes its initial, abrupt response, instead of afterward when barks or continuous barking are more likely. Unlike them, chitter barking is not followed by alertly attentive behaviour.

There is non-behavioural information in all barks, in addition to information about behaviour. They identify the communicator at least as a black-tailed prairie dog and perhaps individually.

Apparent Response to Barks and Barking Patterns

Observations were also made of the behaviour of individuals other than the communicator. After loud, brief barks, in the zoo or the wild, other prairie dogs often ran to burrows and sometimes went directly below ground. Most that remained above turned to face the same stimulus as the communicator. We do not know how many were aware of the stimulus before the barks, but some must have responded

largely or solely did continuous predator other attention to th it and joined i ations in the barking by a f mate coming neighbour. Wh continuous bar of its coterie t ceased approach Chitter barking 'startled' reacti tion of attempt

We must em what extent th these subsequent tions were not available. Nor

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largely or solely to them. When one individual did continuous barking toward a potential predator others gave at least some immediate attention to the stimulus, and sometimes faced it and joined in the barking. In challenge situations in the zoo and Wyoming, continuous barking by a female was often followed by her mate coming and challenging the offending neighbour. When an individual directed barks or continuous barking at a mate or other member of its coterie the latter usually broke contact, ceased approaching, or became more cautious. Chitter barking usually led to an immediate, 'startled' reaction and brief or complete cessation of attempts to interact.

We must emphasize that we do not know to what extent the vocalizations contributed to these subsequent acts by recipients. The vocalizations were not the only sources of information available. Nonetheless, recipients often did

something appropriate to the circumstance, and these vocalizations do provide information that should be relevant.

Other Vocal Displays

Rasp

The rasp is a prolonged, very harsh vocalization (Fig. 3(a)) at times (Fig. 3(b), (c)) similar to prolonged variants of the vocal component of the jump-yip display (Smith et al. 1976). Rasp-like vocalizations are described in the literature, although it is not clear that Seton (1926), Anthony (1955), King (1955), or R. E. Smith (1958) always separated the rasp from what we recognized as harsh chitter barking. Waring's 'snarl' (1970) is comparable to our rasps, although his fainter, somewhat pulsed 'growls' may be intermediate to chitters.

We recognized only 31 rasps in the zoo study, although some were probably missed in the

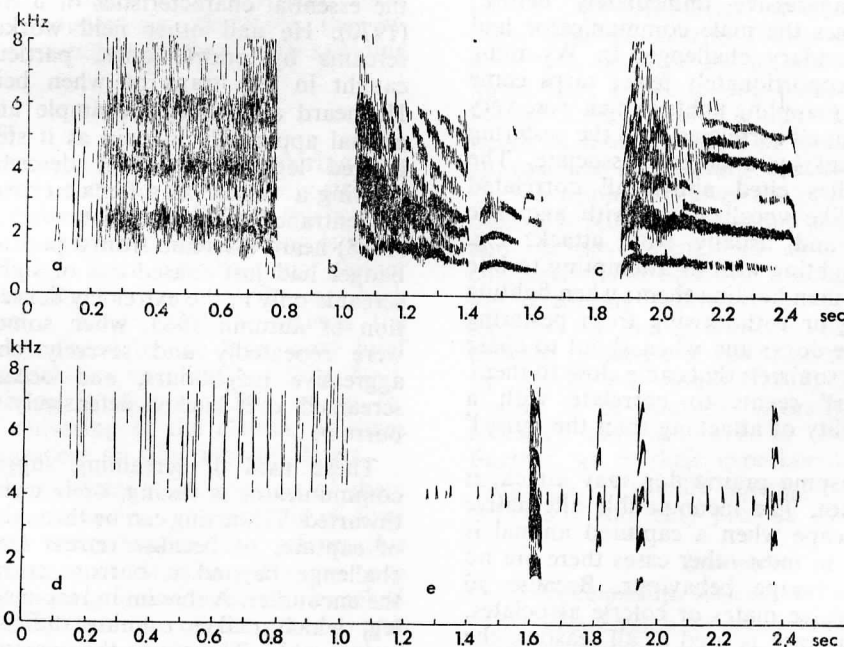


Fig. 3. The rasp (a) an example of moderate duration that shows the noisiness typical of this vocalization, as well as the harmonic structure that is often obscured in recordings. Other rasps may have brief areas without noise that also reveal these harmonics, or may be briefly interrupted by harsh chitter barks. Rasps intergrade with harsh jump-yips having either two (b) or one (c) vocal components. On the lower line a burst of tooth-chattering that we believe to be non-vocal (d) is presented along with a burst we believe to be vocal (e: the large chevrons are barks). Particularly in the former the sonogram reveals little more than vertical lines of noise with slight thickenings indicative of higher amplitudes. The tracings are not accurate in fine detail.

intensely active agonistic encounters during the first year. In the last 2 years, 12 of 14 rasps occurred as or immediately after grappling, biting fights broke up. None of these 12 fights was during a boundary dispute, although 84 (71 per cent) of the 118 fights seen in these 2 years were. Eight of the 12 fights involved pairs, and seven were occasioned as a male approached his mate. In seven of these eight fights the male rasped; in the other his mate called while chasing him off. In the two cases without fighting, one female rasped toward an approaching male neighbour who then did not attack. Another female rasped between two bursts of chitter barking while fleeing from a neighbouring male; they were associates in a coterie that was beginning to subdivide, separating them.

From this correlation with fighting, the rasp appears to indicate that the communicator may attack. In other instances, the individuals who rasped were aggressive immediately before: e.g. in four cases the male communicator had just left a boundary challenge. In Wyoming observations proportionately fewer rasps came during or after grappling fights (which were very uncommon), but all correlated with the pestering of an individual by a coterie associate. The published studies cited above all correlated rasps or rasp-like vocalizations with agonistic circumstances and usually with attack: e.g. uttered when fighting with or attempting to bite a hawk or a human holding them; when fighting with, repulsing or withdrawing from pestering by other prairie dogs; and when about to chase smaller ground squirrels that came close to them. Waring's 'snarl' seems to correlate with a higher probability of attacking than the 'growl' he recognized.

Even if a rasping prairie dog may attack, it usually does not. The incompatible alternative is probably escape when a captured animal is struggling, but in most other cases there are no indications of escape behaviour. Because so many cases involve mates or coterie associates, and because the rasp is used at all seasons, the alternatives could include association and other non-aggressive interaction. Our sample is too small, however, to reveal the non-attack message(s).

Harshness can mark variants of all forms of barks and barking, and the yip of the jump-yip display, although the rasp is the harshest and the only consistently harsh vocalization. Harsh continuous barking or harsh chitter barking were both not uncommonly used (we found a

single harsh bark only once) by a female when approached by her mate or by a neighbouring male. She did not withdraw, but quite rarely fought with him; he usually moved away immediately. On the other hand, when these calls were not harsh he rarely moved away immediately and she might instead. Therefore, although we have little direct evidence that the communicator is likely to attack after using a harsh vocalization, the usual responses of recipients both suggest this, and also make attack unnecessary. In general, audible harshness likely encodes information about attack, whether always linked to a vocalization or facultatively added.

Scream

The scream is clear, high pitched, and always somewhat prolonged. Although we have no recordings, Waring's illustration probably shows the essential characteristics of a brief example (1970). He and other field workers reported screams by prairie dogs, particularly pups, caught in live traps or when being handled. He heard a prolonged example uttered by an animal apparently in pain as it stepped on an injured leg. King (1955) described a male uttering a 'shriek' as a female chased him from the entrance of her nest burrow. R. E. Smith (1958) heard a scream from a prairie dog that a badger had just chased out of sight. We heard screams only in the extremely dense zoo population of autumn 1965, when some individuals were repeatedly and severely challenged by aggressive neighbours, and occasionally one screamed as it backed defensively down into a burrow.

These uses of screaming suggest that the communicator is fleeing, or is trying to, but is thwarted. Thwarting can be through the restraint of capture, or because retreat from a border challenge beyond a burrow entrance forfeits the encounter. A scream in response to pain in a leg, which makes running difficult, could be comparable. Therefore, the scream appears to encode a message of escape, perhaps at high intensity (i.e. escape, if feasible, will be precipitous).

There were no obvious responses to screams in the zoo, but in R. E. Smith's badger case watching prairie dogs 'bolted down their burrows'. King reported that females were attracted by the screams of pups, and sometimes tried to retrieve them.

Yip

The most prominent is the jump-yip which combines a special component so essentially a single the information probable than alternatives that

Males perform apparently become readily to territorial into vacillating frequently seek hand, do about patterned barking monitor but av February and defend small females are com pestered by the interactions. For individuals most interactions.

Tooth Chatter

As the name non-vocal noise together the inc one of three co accompanied muffled barks, pitched whistle noise is even r and variable ba Tooth chattering conspicuous qu the only compo

We do not u acts to display usage suggests functions. For i seen solitary r over and holdin very quietly cha cold. Cates (192 chattered quietl 'keeping the t prairie dogs ha allogroomed (R by a human (V our studies and (1955) the most tooth chattering

Yip

The most prominent display of the species is the jump-yip (Smith et al. 1976). It combines a specialized leap with a vocalization, components so rarely separated that they are essentially a single pattern. The pattern encodes the information that escape behaviour is less probable than any of a general set of unspecified alternatives that are incompatible with escape.

Males perform 75 per cent of the jump-yips, apparently because they seek and respond readily to territorial challenging, thus getting into vacillating situations from which they frequently seek escape. Females, on the other hand, do about 30 per cent of all barks and patterned barking. In contrast to males, they monitor but avoid most challenges (except in February and sometimes March when they defend small nest territories). Further, when females are coming into heat they are much pestered by the males, and also shun these interactions. Females are therefore the individuals most usually concerned with avoiding interactions.

Tooth Chatter

As the name suggests, tooth chattering is a non-vocal noise produced by repeatedly striking together the incisors (Fig. 3(d)), but this is only one of three components. The striking is often accompanied by faint vocalizations (either muffled barks, R. E. Smith 1958; or high pitched whistles, King 1955). Sometimes the noise is even replaced by the faint, very brief and variable bark-like vocalizations (Fig. 3(e)). Tooth chattering is always visible as an inconspicuous quivering of the cheeks, at times the only component an observer can detect.

We do not understand the relation of these acts to display behaviour, as not all of their usage suggests specialization for informative functions. For instance, on wintry days we have seen solitary resting individuals sit, hunched over and holding their feet off the ground, and very quietly chattering their teeth. They looked cold. Cates (1927) noted that his pet prairie dog chattered quietly even on warm days, perhaps 'keeping the teeth sharpened'. Other resting prairie dogs have tooth chattered while being allogroomed (R. E. Smith 1958) or groomed by a human (Waring's pet, 1970). Yet in both our studies and King's prolonged observations (1955) the most prominent and abundant use of tooth chattering was in agonistic interactions.

We logged 757 bouts of tooth chatter in our zoo studies, but some were probably overlooked, and others that repeatedly waxed and waned may have been tallied more than once. We were able to assign 689 of the 757 to agonistic categories. In all agonistic cases the tooth chattering individual oriented toward the source of trouble, rather than away or toward an escape route. Individuals engaged in boundary challenges (511 cases) or watching such events (47 cases) accounted for 81 per cent of the 689. In another 108 an animal tooth chattered while watching some potential source of danger such as a peafowl or human. In 23 cases an individual used tooth chatter in response to attempts by another individual to interact: a female to her mate in the breeding or pre-breeding period, sometimes the male on being rebuffed (they might then fight), and sometimes males interacting at other seasons.

King (1955) described tooth chatter as characteristic of 'disputing' (challenging) prairie dogs, although Waring (1970) sought and was never convinced he found such use. King also noted that a prairie dog would tooth chatter toward 'some object of which it is not afraid but which it does not like'; he even records being approached by a tooth chattering individual. On the whole, whatever the possible non-signal uses of striking the incisors together, what King and we have recognized as tooth chatter is characteristic of agonistic situations from which the chattering individual is not immediately likely to flee and in which it sometimes presses forward in attack or incipient attack. A survey by Brooks & Banks (1973) revealed tooth chatter to be used by 'virtually all species of rodents', although not by other mammals. Further, as in their experimental results with lemmings, 'it usually appears related to threat behaviour' and may precede attack.

Discussion

Form Relationships Among the Vocalizations

No vocalization of black-tailed prairie dogs is either invariant or fully discrete. Rasps vary in harshness, and at their least harsh are closely similar to relatively long and harsh yips recorded occasionally in jump-yip performances. Much briefer yips intergrade with both broadly peaked and bipeaked barks. Barks and continuous barking intergrade when series have irregular intervals. Barks and chitter barking intergrade when chitter bursts comprise only two bark units. Chitter barking and continuous

barking intergrade both when continuous series of barks briefly become sharply peaked and very rapid, and when short bursts become repeated at irregular intervals (thus far known only for bursts comprising broadly peaked barks). Mews are very incompletely understood, but appear to intergrade with continuous barking in which the units peak at low frequencies. And the apparently vocal forms of tooth chattering resemble very rapid, very brief unit continuous barking. Screams vary in at least duration, but may not intergrade with other calls; they have rarely been tape recorded.

As an alternative to describing the repertoire in terms just of a web of interrelated vocalizations, it is possible to expand the catalogue and list the different kinds of specializations of form that have their own messages. At least eight are identified above: the prolonged rasp vocalization, harshness in it or in other vocalizations, the range of yip forms, the briefer bark form, the yet briefer sharply peaked bark, the regular pattern of repetition in continuous barking, the compact burst arrangement of chitter barking, and the staccato repetition in tooth chattering. In addition, screams need to be further characterized, and other specializations may exist, particularly if there are faint vocalizations not heard on the surface.

Most or all of the messages encoded by the vocalizations of black-tailed prairie dogs appear to be encoded by displays of many other species of at least vertebrate animals. Among messages specifying selections from the behavioural repertoire of the communicator, escape and attack are extremely widespread, as is the encoding of the information that a communicator may select an otherwise unspecified alternative that is incompatible with some selection (termed the 'general set message', W. J. Smith 1969). A message of non-specific interacting behaviour is at present known only from this ground squirrel, from a display of gorillas and humans (Smith, Chase & Leiblich 1974), and from at least one bird species (*Myiozetetes similis*, W. J. Smith, in press); it may also be widespread, at least among persistently social species. The message specifying the behaviour of being attentive has been recognized only recently, and although it or some messages very similar to it are found in displays of other species, the extent of its distribution is unknown.

Messages about behaviour provide information about more than the kinds of acts that may be selected; they also provide supplementary

information about these selections, e.g. the probability that a particular selection will occur. Information about probability is universally encoded. Information about the relative stability of probabilities is universally encoded in tonic signalling patterns, and these appear to be widespread. Some displays may provide information about the intensity with which acts will be performed, e.g. escape may be abrupt or gradual. Finally, among categories of non-behavioural messages, information about at least species identity is nearly universal in vocalizations.

The vocalizations of black-tailed prairie dogs thus do not encode kinds of information peculiar to this species. This appears to be typical, at least among vertebrates (W. J. Smith 1969). Yet kinds of information that are particularly important to the prairie dogs are made readily available: the likelihood of escape, for instance, and the activity of being persistently attentive while rejecting interaction.

Messages are also encoded by this species in display behaviour other than vocalizations. Those encoded in visible and tactile patterns and the use of pheromones will be treated in a further paper.

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REFERENCES

- Anthony, A. (1955). Behavior patterns in a laboratory colony of prairie dogs, *Cynomys ludovicianus*. *J. Mammal.*, 36, 69-78.
- Brooks, R. J. & Banks, E. M. (1973). Behavioural biology of the collared lemming [*Dicrostonyx groenlandicus* (Traill)]: an analysis of acoustic communication. *Anim. Behav. Monogr.*, 6, 1-83.
- Cates, E. C. (1927). Notes concerning a captive prairie dog. *J. Mammal.*, 8, 33-37.
- King, J. A. (1955). and popul dog town Univ. M. 1-123.
- Schleidt, W. M. (effects of systems. J.
- Seton, E. T. (19 vicianus) 229-230.
- Smith, R. E. (19 in Kansas State Biol.
- Smith, W. J. (19 cation. Sc

- King, J. A. (1955). Social behavior, social organization, and population dynamics in a black-tailed prairie dog town in the Black Hills of South Dakota. *Univ. Michigan, Contr. Lab. Vert. Biol.*, **67**, 1-123.
- Schleidt, W. M. (1973). Tonic communication: continual effects of discrete signs in animal communication systems. *J. Theor. Biol.*, **42**, 359-386.
- Seton, E. T. (1926). The prairie-dogs (*Cynomys ludovicianus*) at Washington zoo. *J. Mammal.*, **7**, 229-230.
- Smith, R. E. (1958). Natural history of the prairie dog in Kansas. *Univ. Kans. Mus. nat. Hist. and State Biol. Surv. Misc. Publ.*, **16**, 1-36.
- Smith, W. J. (1969). Messages of vertebrate communication. *Science, N.Y.*, **165**, 145-150.
- Smith, W. J., Chase, J. & Leiblich, A. K. (1974). Tongue showing: a facial display of humans and other primate species. *Semiotica*, **11**, 201-246.
- Smith, W. J., Smith, S. L., Oppenheimer, E. L., deVilla, J. G. & Ulmer, F. A. (1973). Behaviour of a captive population of black-tailed prairie dogs. Annual cycle of social behaviour. *Behaviour*, **46**, 189-220.
- Smith, W. J., Smith, S. L., deVilla, J. G. & Oppenheimer, E. L. (1976). The jump-yip display of the black-tailed prairie dog *Cynomys ludovicianus*. *Anim. Behav.*, **24**, 609-621.
- Waring, G. H. (1970). Sound communications of black-tailed, white-tailed, and Gunnison's prairie dogs. *Am. Midl. Nat.*, **83**, 167-185.

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